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Toripalimab Plus Chemotherapy for Recurrent or Metastatic Nasopharyngeal Carcinoma
The JUPITER-02 Randomized Clinical Trial

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IMPORTANCE There are currently no therapies approved by the US Food and Drug Administration for nasopharyngeal carcinoma (NPC). Gemcitabine-cisplatin is the current standard of care for the first-line treatment of recurrent or metastatic NPC (RM-NPC).

OBJECTIVE To determine whether toripalimab in combination with gemcitabine-cisplatin will significantly improve progression-free survival and overall survival as first-line treatment for RM-NPC, compared with gemcitabine-cisplatin alone.

DESIGN, SETTING, AND PARTICIPANTS JUPITER-02 is an international, multicenter, randomized, double-blind phase 3 study conducted in NPC-endemic regions, including mainland China, Taiwan, and Singapore. From November 10, 2018, to October 20, 2019, 289 patients with RM-NPC with no prior systemic chemotherapy in the RM setting were enrolled from 35 participating centers.

INTERVENTIONS Patients were randomized (1:1) to receive toripalimab (240 mg [n = 146]) or placebo (n = 143) in combination with gemcitabine-cisplatin for up to 6 cycles, followed by maintenance with toripalimab or placebo until disease progression, intolerable toxicity, or completion of 2 years of treatment.

MAIN OUTCOME Progression-free survival as assessed by a blinded independent central review. Secondary end points included objective response rate, overall survival, progression-free survival assessed by investigator, duration of response, and safety.

RESULTS Among the 289 patients enrolled (median age, 46 [IQR, 38-53 years; 17% female), at the final progression-free survival analysis, toripalimab treatment had a significantly longer progression-free survival than placebo (median, 21.4 vs 8.2 months; HR, 0.52 [95% CI, 0.37-0.73]). With a median survival follow-up of 36.0 months, a significant improvement in overall survival was identified with toripalimab over placebo (hazard ratio [HR], 0.63 [95% CI, 0.45-0.89]; 2-sided P = .008). The median overall survival was not reached in the toripalimab group, while it was 33.7 months in the placebo group. A consistent effect on overall survival, favoring toripalimab, was found in subgroups with high and low PD-L1 (programmed death–ligand 1) expression. The incidence of all adverse events, grade 3 or greater adverse events, and fatal adverse events were similar between the 2 groups. However, adverse events leading to discontinuation of toripalimab or placebo (11.6% vs 4.9%), immune-related adverse events (54.1% vs 21.7%), and grade 3 or greater immune-related adverse events (9.6% vs 1.4%) were more frequent in the toripalimab group.

CONCLUSIONS AND RELEVANCE The addition of toripalimab to chemotherapy as first-line treatment for RM-NPC provided statistically significant and clinically meaningful progression-free survival and overall survival benefits compared with chemotherapy alone, with a manageable safety profile. These findings support the use of toripalimab plus gemcitabine-cisplatin as the new standard of care for this patient population.

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Visual Abstract

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Supplemental content

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Nasopharyngeal carcinoma (NPC) is a rare form of head and neck cancer[1] in the West, with an annual incidence of less than 1 per 100 000 individuals in the US, while it is much more prevalent in Southern China and Southeast Asia, with an annual incidence of about 9 per 100 000.[2,3] Latent infection with Epstein-Barr virus (EBV) is crucial for the development of NPC,[4] with environmental factors such as tobacco and alcohol consumption playing lesser roles.[5,6]

No drugs have been approved by the US or European health authorities for the treatment of NPC. Gemcitabine-cisplatin chemotherapy is the current standard first-line treatment for patients with recurrent or metastatic NPC (RM-NPC), as recommended by National Comprehensive Cancer Network and European Society for Medical Oncology guidelines.[7-9] Three recent randomized phase 3 trials—JUPITER-02, CAPTAIN-1st, and RATIONALE-309—have reported similar significant improvements in progression-free survival with the addition of a programmed cell death protein 1 (PD-1)-blocking antibody to the gemcitabine-cisplatin backbone for the first-line treatment of RM-NPC.[10-12] However, due to the immaturity of survival data, a definitive analysis of survival benefit has not yet been reported.

Toripalimab, a humanized IgG4K monoclonal antibody against human PD-1,[13-16] was approved as a first-line treatment for RM-NPC in November 2021 in China. The approval was based on the results of JUPITER-02, a randomized phase 3 study comparing gemcitabine-cisplatin in combination with toripalimab vs placebo in patients with RM-NPC. At the interim progression-free survival analysis, a significant improvement in progression-free survival was detected for toripalimab over placebo.[10] Here we report results from the prespecified definitive overall surival analysis.

Key Points

Question Will an immune checkpoint blocker in combination with the first-line chemotherapy improve progression-free survival and overall survival in patients with recurrent or metastatic nasopharyngeal carcinoma?

Findings At the final progression-free survival analysis, toripalimab treatment had a significantly longer progression-free survival than placebo. The prespecified final overall survival analysis revealed that the addition of toripalimab to gemcitabine-cisplatin led to statistically significant and clinically meaningful improvement in survival compared with chemotherapy alone, with a manageable safety profile.

Meaning The demonstrated progression-free survival and overall survival benefits support the use of toripalimab in combination with gemcitabine-cisplatin as the new standard first-line treatment for patients with recurrent or metastatic nasopharyngeal carcinoma.

Methods

Patients

Patients eligible for this study were between the ages of 18 and 75 years and were required to have histologically or cytologically confirmed primary RM-NPC, which was not amenable for local-regional or curative treatment, and must have received no prior systemic chemotherapy in the RM setting. The full eligibility criteria are available in the study protocol (Supplement 1).

Trial Design and Treatment

Patients were randomly assigned in a 1:1 ratio to receive either toripalimab or placebo in combination with gemcitabine and cisplatin once every 3 weeks for up to 6 cycles, followed by toripalimab or placebo maintenance. Stratification factors included tumor status (recurrent vs primary metastatic) and Eastern Cooperative Oncology Group performance status score (0 vs 1). During the chemotherapy phase, patients would receive toripalimab (240 mg) or placebo on day 1, gemcitabine (1000 mg per square meter of body-surface area) on days 1 and 8, and cisplatin (80 mg/m^2) on day 1 of each 3-week cycle. During the maintenance phase, patients would receive toripalimab (240 mg) or placebo once every 3 weeks until progressive disease, intolerable toxicity, withdrawal of consent, or a maximum of 2 years of treatment. The detailed randomization method can be found in in Supplement 1 and Supplement 2.

End Points

The primary end point was progression-free survival in the intention-to-treat population as assessed by a blinded independent central review (BICR) per response evaluation criteria in solid tumors (RECIST) version 1.1. Secondary end points included objective response rate (ORR), overall survival, progression-free survival assessed by investigator, duration of response, and safety. Overall survival was defined as the time from randomization to death from any cause. Progression-free survival was defined as the time from randomization to the time of first documented disease progression or death. The complete list of end points and statistical analysis details can be found in the protocol (Supplement 1) and statistical analysis plan (Supplement 2).

PD-L1 Expression

Archival or fresh tumor biopsy samples were obtained from patients prior to treatment. Expression of programmed death ligand 1 (PD-L1) was evaluated by immunohistochemistry staining using a validated JS311 assay on the Ventana Benchmark Ultra platform in a central laboratory.[17] PD-L1 positivity was defined as the presence of membrane staining of any intensity in 1% or greater of tumor cells or immune cells.

Trial Oversight

The trial protocol and amendments were approved by the institutional review board or ethics committee at each participating site. All patients provided written informed consent at enrollment. This trial was conducted in full conformance with the International Council for Harmonisation E6 guideline for Good Clinical Practice and the principles of the Declaration of Helsinki. The independent data-monitoring committee periodically assessed the safety data and reviewed the efficacy results at the interim progression-free

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survival analysis. All analyses for the independent data-monitoring committee review were prepared by an independent third party.

Statistical Analysis

The sample size calculation was based on BICR-assessed progression-free survival, the primary end point. A total of 280 patients were needed to observe 200 progression-free survival events to detect a progression-free survival improvement of hazard ratio (HR) = 0.67 with 80% power at a 2-sided significance level of .05. An interim progression-free survival analysis was planned at 130 progression-free survival events. O'Brien-Fleming boundaries were used to control the overall type 1 error rate for progression-free survival interim and final analyses. The estimated median overall survival for gemcitabine-cisplatin chemotherapy was 29 months. The definitive final overall survival analysis was planned at 130 deaths with 46% power to detect an overall survival improvement of 28% risk reduction in death (HR, 0.72). Once the null hypothesis of the primary end point was rejected, the key secondary end points, BICR-assessed ORR and overall survival, were to be tested hierarchically under a 2-sided α level of .05. Each overall survival analysis conducted before the definitive analysis, including ad hoc analysis requested by regulatory health authorities, assumed a nominal α of 1e-6. The efficacy boundary for the final overall survival analysis was .049995.

The 2-sided log-rank test, stratified by baseline Eastern Cooperative Oncology Group performance status score (0 vs 1) and baseline disease stage (recurrent vs primary metastatic), was used as the primary analysis to compare progression-free survival and overall survival between the 2 treatment groups. The hazard ratio was estimated with the use of the stratified Cox proportional hazards model. The Kaplan-Meier methodology was applied to estimate the median progression-free survival and overall survival for each treatment group. The Brookmeyer Crowley methodology was used to construct the 95% CI for the median progression-free survival and overall survival for each treatment group.

Efficacy was assessed in the intention-to-treat population (n = 289), which included all randomized patients. Safety analyses were performed on the safety population (n = 289), which included all randomized patients who received any amount of study drug. The survival follow-up cutoff date was November 18, 2022, while the safety cutoff date was May 8, 2022 (7 months after the last patient completed study treatment).

Results

Patients and Treatment

From November 10, 2018, to October 20, 2019, 35 participating centers from mainland China, Taiwan, and Singapore screened 408 patients with RM-NPC, and 289 were randomized 1:1 to either the toripalimab group (n = 146) or the placebo group (n = 143) (**Figure 1**; eTable 1 in Supplement 3). Patients who experienced disease progression more than 6 months after definitive therapy with curative intent were allowed to be enrolled.

The primary reasons for screen failure were not meeting eligibility criteria and withdrawal of consent (Figure 1; eTable 2 in Supplement 3). The median age was 46 (IQR, 38-53) years in the toripalimab group and 51 (IQR, 43-57) years in the placebo group; 85% and 81% patients were male in the toripalimab and placebo groups, respectively. The baseline demographic and characteristics were generally balanced between the 2 groups, with the exception that the median age was 5 years younger in the toripalimab group relative to the placebo group (**Table 1**). Fresh or archival tumor biopsy samples were obtained for evaluation of PD-L1 expression (eTable 3 in Supplement 3).

Both groups received a median of 6 cycles of chemotherapy, and approximately 80% of all patients continued to receive toripalimab or placebo maintenance treatment (Figure 1; eTable 4 in Supplement 3). By October 2021, all remaining patients completed the protocol-defined 2-year maximum of study treatment (including 36% of patients from the toripalimab group) and continued in survival follow-up (Figure 1). By November 18, 2022, 82 of 146 patients (56.2%) from the toripalimab group and 106 of 143 (74.1%) from the placebo group received additional anticancer therapies after the study treatment (eTable 5 in Supplement 3), including 48 of 146 (33%) from the toripalimab group and 49 of 143 (34%) from the placebo group who received later line anti-PD-1/PD-L1 treatments (eTable 5 in Supplement 3).

Progression-Free Survival

Because the interim progression-free survival results had crossed the efficacy boundary, the prespecified final progression-free survival analysis was thus descriptive. By the cutoff date of June 8, 2021, toripalimab treatment extended the median progression-free survival by 13.2 months over the placebo as assessed by BICR (21.4 vs 8.2 months). The HR was 0.52 (95% CI, 0.37-0.73; nominal $P < .001$) (**Figure 2** and **Table 2**). The 1- and 2-year progression-free survival rates were 59.0% vs 32.9% and 44.8% vs 25.4% in the toripalimab and placebo groups, respectively. Consistent progression-free survival treatment effects were observed across all major subgroups (eFigure 1 in Supplement). A similar improvement in progression-free survival was observed as assessed by investigators (eFigure 4 in Supplement 3).

Overall Survival

As of November 18, 2022, with a median survival follow-up of 36.0 months, 133 overall survival events (57 from the toripalimab group and 76 from the placebo group) were recorded, triggering the prespecified final overall survival analysis (eTable 6 in Supplement 3). Among surviving patients, the median follow-up was 39.6 (range, 1.2-47.4) months for the toripalimab group and 39.9 (range, 0.2-48.3) months for the placebo group. The median overall survival was not reached in the toripalimab group, while it was 33.7 months in the placebo group. A significant improvement in overall survival was detected for toripalimab over placebo (HR, 0.63 [95% CI, 0.45-0.89]; 2-sided $P = .008$, crossing the prespecified efficacy

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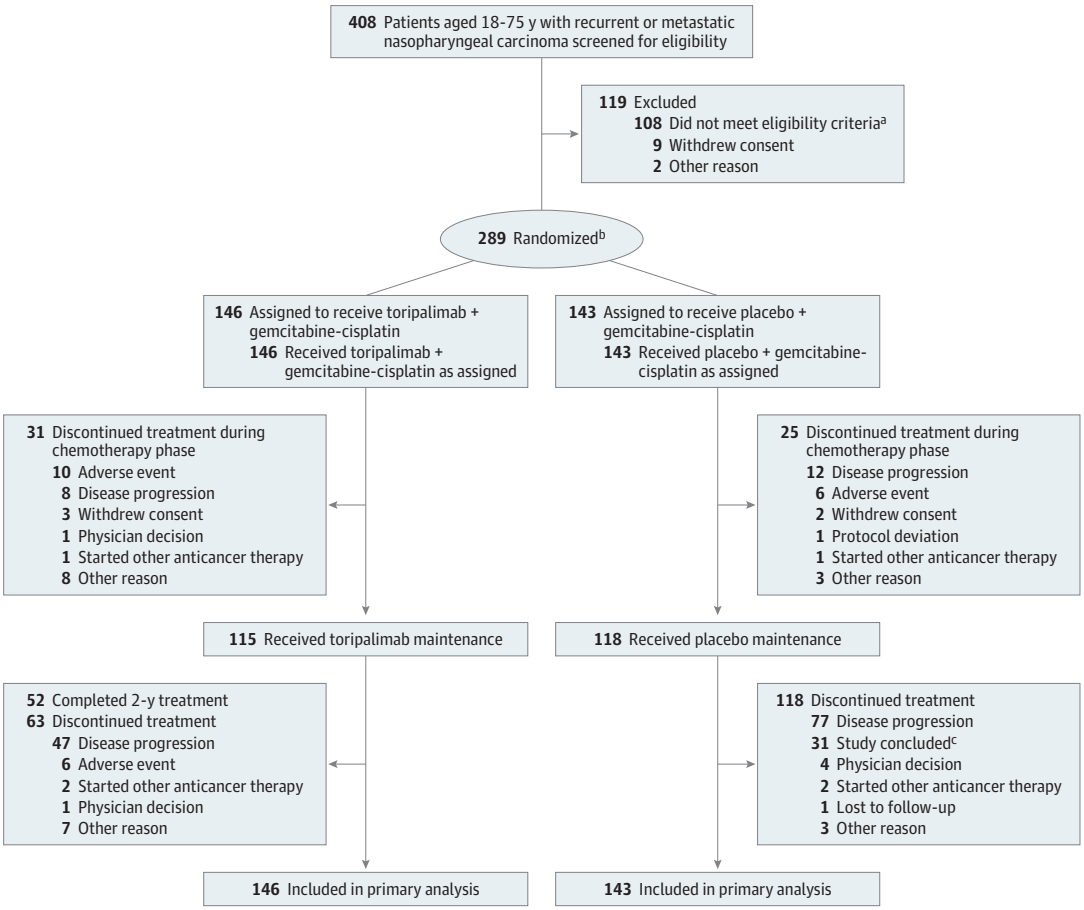
Figure 1. Screening, Randomization, and Flow of Patients in a Trial of Toripalimab for Nasopharyngeal Carcinoma



[a] See eTable 2 in Supplement 3 for details about not meeting eligibility criteria.

[b] Patients were randomized in a 1:1 ratio, stratified by baseline Eastern Cooperative Oncology Group performance status score (0 vs 1) and baseline disease stage (recurrent vs primary metastatic). Primary metastatic disease was defined as de novo metastatic disease at the initial diagnosis.

[c] A prespecified interim analysis of progression-free survival was conducted by the data cutoff date of May 30, 2020. Because the interim progression-free survival results showed a significant improvement of progression-free survival in the toripalimab group, the independent data-monitoring committee recommended to unblind the study. Patients in the toripalimab group continued to receive toripalimab treatment, whereas patients in the placebo group (n = 31) discontinued placebo and concluded the study.

boundary) (Figure 2 and Table 2). At 1 year, the overall survival rates were 90.9% in the toripalimab group vs 87.1% in the placebo group; at 2 years, 78.0% vs 65.1%; and at 3 years, 64.5% vs 49.2%. The overall survival treatment effects were generally consistent across major subgroups, with the exception of patients with primary metastatic disease at baseline (eFigure 1 in Supplement 3). Notably, the treatment effects on overall survival favored toripalimab in all PD-L1 expression subgroups and EBV DNA copy number subgroups (eTable 7 and eFigures 2-3 in Supplement 3).

Tumor Response

By the cutoff date of June 8, 2021, the BICR-assessed ORRs were 78.8% vs 67.1% in the 2 groups. The complete response rates as assessed by BICR in the toripalimab group were twice as high as in the placebo group: 26.7% vs 13.3% (Table 2). The responses were durable, as the median duration of response assessed by BICR was 18.0 months (95% CI, 10.5 to not estimable) in the toripalimab group and 6.0 months (95% CI, 5.6-8.3) in the placebo group (HR, 0.49 [95% CI, 0.33-0.72]) (Table 2; eFigure 5 in Supplement 3).

EBV DNA Copy Number

Dynamic plasma EBV DNA copy number was monitored for correlation with clinical response (Table 2; eFigure 6 in Supplement 3). Among patients with detectable EBV DNA copy number at baseline and at least 1 posttreatment EBV result, 107 of 107 patients in the toripalimab group and 99 of 103 (96.1%) in the placebo group experienced EBV DNA copy number reduction after the study treatments, including 103 of 107 (96.3%) in the toripalimab group and 87 of 103 (84.5%) in the placebo group who had EBV DNA copy number decreased to undetectable level. After the initial reduction, 39 of 107 patients (36.5%) in the toripalimab group and 58 of 101 (57.4%) in the placebo group experienced EBV DNA copy number rebound. The median time from the lowest EBV DNA

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Table 1. Baseline Patient Characteristics

Characteristic	No. (%)	
	Toripalimab + gemcitabine-cisplatin (n = 146)	Placebo + gemcitabine-cisplatin (n = 143)
Age, median (IQR), y	46 (38-53)	51 (43-57)
≤50	97 (66)	69 (48)
>50	49 (34)	74 (52)
Sex		
Male	124 (85)	116 (81)
Female	22 (15)	27 (19)
Asian	146 (100)	143 (100)
Disease status		
Recurrent	85 (58)	87 (61)
Locally advanced, No./total (%)	19/85 (22)	20/87 (23)
With distal metastasis, No./total (%)	66/85 (78)	67/87 (77)
Primary metastatic[a]	61 (42)	56 (39)
ECOG PS score[b]		
0	83 (57)	81 (57)
1	63 (43)	62 (43)
Histology		
Nonkeratinizing squamous cell carcinoma	145 (99)	140 (98)
Keratinizing squamous cell carcinoma	1 (1)	2 (1)
Other	0	1 (1)
Metastatic sites at baseline (ITT population)		
Liver	61 (42)	57 (40)
Bone	60 (41)	55 (39)
Lung	59 (40)	56 (39)
Current or former smoker	76 (52)	59 (41)
Current or former alcohol use	30 (21)	18 (13)
Prior treatment		
Radiation therapy	85 (58)	87 (61)
Concurrent chemotherapy to radiation	65 (45)	68 (48)
Neoadjuvant therapy	54 (37)	49 (34)
Surgery	38 (26)	43 (30)
Adjuvant therapy	24 (16)	21 (15)
Disease-free interval, y, No./total (%)[c]		
≤2	48/75 (64)	45/79 (57)
>2	27/75 (36)	34/79 (43)
PD-L1 status positive, No./total (%)[d]	109/130 (84)	109/133 (82)
Baseline plasma EBV DNA copy No., IU/mL[e]		
<2000	54 (37)	54 (38)
≥2000	92 (63)	89 (62)

Abbreviations: ECOG PS, Eastern Cooperative Oncology Group performance status; ITT, intention to treat; PD-L1, programmed cell death ligand 1; EBV, Epstein-Barr virus.

[a] Primary metastatic disease is defined as de novo metastatic disease at the initial diagnosis.

[b] Scores range from 0 to 5, with 0 indicating no symptoms and higher scores indicating greater disability: 0, asymptomatic; 1, symptomatic but completely ambulatory; 2, symptomatic, in bed less than 50% during the day; 3, symptomatic, in bed more than 50% during the day but not bedbound; 4, bedbound; 5, death.

[c] Evaluated among patients who had received definitive radiation or chemoradiation therapy as the last treatment before enrolling onto the current study. The disease-free interval was calculated from the end of definitive therapy to disease progression prior to the study treatments.

[d] Defined as 1% or more of tumor cells or 1% or more of immune cells expressing PD-L1 by JS311 IHC staining in a central laboratory.

[e] Determined by real-time quantitative reverse transcription polymerase chain reaction method with probes against EBV genes in a central laboratory.

copy number to the rebound was 20.5 vs 6.0 months in the toripalimab and placebo groups, respectively. The rebound also preceded investigator-assessed disease progression by a median of 1.9 months in the toripalimab group (Table 2).

Adverse Events

By the safety cutoff date of May 8, 2022, the median treatment duration was 15.3 months in the toripalimab group and 8.7 months in the placebo group (eTable 4 in Supplement 3). All patients experienced at least 1 treatment-emergent adverse event (TEAE) (**Table 3**). The overall safety profile was consistent with the interim report and no new signals emerged after more extensive toripalimab exposure. The overall incidences of grade 3 or higher TEAEs (89.7% vs 90.2%), fatal TEAEs (3.4% vs 2.8%), serious adverse events (43.8% vs 43.4%), and infusion reactions (4.1% vs 4.2%) were

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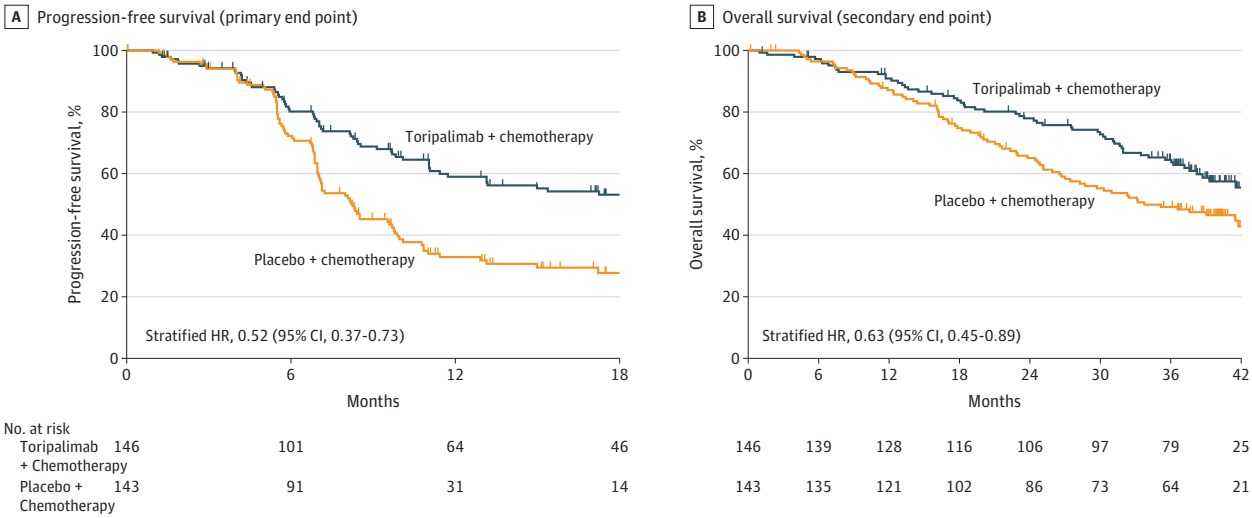
Figure 2. Final Progression-Free Survival and Overall Survival Analyses in the Intention-to-Treat Population



A, Estimated progression-free survival as assessed by blinded independent central review according to RECIST (Response Evaluation Criteria in Solid Tumors) version 1.1. Median progression-free survival was 21.4 (IQR, 7.1 to not estimable) months in the toripalimab group and was 8.2 (IQR, 5.7 to not estimable) months in the placebo group. B, The median overall survival was not reached (IQR, 27.6 months to not estimable) in the toripalimab group and was 33.7 (IQR, 17.8 to not estimable) months in the placebo group. Vertical ticks on curves indicate censored patients. Hazard ratios were stratified by baseline Eastern Cooperative Oncology Group performance status score (0 vs 1) and baseline disease stage (recurrent vs primary metastatic).

similar between the 2 groups (eTable 8 in Supplement 3). The most common TEAEs and most common grade 3 or higher TEAEs were dominated by chemotherapy-induced toxicity (Table 3; eTables 9-10 in Supplement 3). TEAEs that occurred more often in the toripalimab group than in the placebo group included hypothyroidism (36.3% vs 17.5%), upper respiratory tract infection (25.3% vs 14.0%), and pneumonia (17.8% vs 7.0%) (eTable 9 in Supplement 3).

As expected, immune-related adverse events determined by the investigators (54.1% vs 21.7%) and immune-related adverse events grade 3 or greater (9.6% vs 1.4%) were more frequent in the toripalimab group (eTables 11-12 in Supplement 3). No novel safety signals were identified when toripalimab compared with other drugs in the same class.

Discussion

This randomized phase 3 study aimed to compare toripalimab and placebo, when given in combination with first-line chemotherapy, for the treatment of RM-NPC. The study results showed that the addition of toripalimab to the standard platinum doublet chemotherapy resulted in significantly better progression-free survival and overall survival than chemotherapy alone, while maintaining a manageable safety profile.

Three recent phase 3 trials—JUPITER-02, CAPTAIN-1st, and RATIONALE-309—evaluated the combination of PD-1 blockade with gemcitabine-cisplatin for the first-line treatment of RM-NPC in NPC-endemic regions. All 3 trials set progression-free survival as the primary end point, while only JUPITER-02 had overall survival as a secondary end point with formal testing and type I error controlled. Collectively, the results from the 3 trials provided strong evidence for improved progression-free survival with the addition of a PD-1 blocking monoclonal antibody to gemcitabine-cisplatin as first-line treatment for RM-NPC. More importantly, data from a 3-year survival follow-up of JUPITER-02 revealed that toripalimab plus gemcitabine-cisplatin demonstrated significant overall survival benefits over gemcitabine-cisplatin alone. Toripalimab reduced the risk of death by 37% (HR, 0.63 [95% CI, 0.45-0.89]; P = .008), crossing the prespecified efficacy boundary.

The significant improvement in overall survival was achieved despite extensive crossover to immunotherapy. At the interim progression-free survival analysis by the cut-off date of May 30, 2020, we reported a low crossover, with only 2 patients in the toripalimab group and 4 in the placebo group receiving later-line anti-PD-1/PD-L1 therapy. However, the later-line treatment paradigm for RM-NPC rapidly evolved in just a few years. Toripalimab monotherapy became the first immune checkpoint inhibitor to receive conditional approval in China as a third-line treatment for RM-NPC in February 2021. The approval was based on the results from a single-group, phase 2 study (POLARIS-02).[15] In POLARIS-02, toripalimab demonstrated clinical benefit in chemorefractory NPC, with an ORR of 20.5%, a median duration of response of 12.8 months, and a median overall survival of 17.4 months.[15] Immune checkpoint inhibitor therapy was quickly adopted as an essential component for the clinical management of RM-NPC after chemotherapy. At the final overall survival analysis, for patients who had experienced disease progression, 48 of 104 (46%) in the toripalimab group and 49 of 123 (40%) in the control group received later-line

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Table 2. Comparisons of Tumor Response by Blinded Independent Central Review per RECIST v1.1, and plasma EBV DNA copy number results in JUPITER-02 study

	Toripalimab + gemcitabine-cisplatin	Placebo + gemcitabine-cisplatin	Difference or HR (95% CI)	*P* value
Progression-free survival by blinded independent central review per RECIST v1.1 (primary end point)				
Progression-free survival, median (95% CI), month	21.4 (11.7 to NE)[a]	8.2 (7.0 to 9.8)[a]	HR, 0.52 (0.37-0.73)[b]	<.001[c]
Overall survival (secondary end point)				
Overall survival, median (95% CI), mo	NE (38.7 to NE)[a]	33.7 (27.0 to 44.2)[a]	HR, 0.63 (0.45-0.89)[b]	.008[c]
Tumor response by blinded independent central review per RECIST v1.1 (secondary end points)				
Objective response rate (complete response + partial response), No./total (%) [95% CI]	115/146 (78.8) [71.2-85.1][d]	96/143 (67.1) [58.8-74.8][d]	Difference, 11.4 [1.7-21.2][e]	.02[f]
Disease control rate (complete response + partial response + stable disease), No./total (%) [95% CI]	129/146 (88.4) [82.0-93.1][d]	115/143 (80.4) [73.0-86.6][d]	Difference, 7.9 [−0.4 to 16.1][e]	.06[f]
Duration of response, median (95% CI), mo	18.0 (10.5 to NE)[a] [n = 115]	6.0 (5.6-8.3)[a] [n = 96]	HR, 0.49 (0.33-0.72)[g]	<.001[h]
Best overall response (response components), No./total (%)				
Complete response	39/146 (26.7)	19/143 (13.3)		
Partial response	76/146 (52.1)	77/143 (53.8)		
Stable disease	14/146 (9.6)	19/143 (13.3)		
Progressive disease	5/146 (3.4)	8/143 (5.6)		
Not evaluable	9/146 (6.2)	8/143 (5.6)		
Non-complete response/ non-progressive disease	3/146 (2.1)	11/143 (7.7)		
No evidence of disease at entry	0	1/143 (0.7)		
EBV DNA copy No. (exploratory end points), No./total (%)				
Decreased from baseline	107/107 (100)	99/103 (96.1)	Difference, 3.9 (0.3-9.6)[i]	.06[j]
Became undetectable after treatments	103/107 (96.3)	87/103 (84.5)	Difference, 11.8 (4.1-20.5)[i]	.004[j]
Rebounded after the initial reduction	39/107 (36.5)	58/103 (57.4)	Difference, −21.0 (−33.8 to −7.4)[i]	.002[j]
Time from the lowest EBV copy No. to rebound, median (IQR), mo	20.5 (9.2 to NE) [n = 107]	6.0 (4.1-14.3) [n = 99]	HR, 0.35 (0.23-0.53)[g]	<.001[h]
Time from EBV copy No. rebound to investigator-identified progressive disease, median (IQR), mo	1.9 (0-4.2) [n = 33]	1.3 (0-2.8) [n = 45]	Difference, 0.6 (−1.1 to 2.3)[k]	.50[k]

Abbreviations: EBV, Epstein-Barr virus; NE, not estimable; RECIST, Response Evaluation Criteria in Solid Tumors.

Data cutoff date: November 18, 2022, for overall survival, and June 8, 2021, for all other results.

[a] The confidence interval for the median survival was computed using the Brookmeyer-Crowley method with log-log transformation.

[b] Computed from the Cox proportional hazards regression model, stratified by the baseline Eastern Cooperative Oncology Group performance status (0 vs 1) and the baseline disease stage (recurrent vs primary metastatic).

[c] Computed from the log-rank test, stratified by the baseline Eastern Cooperative Oncology Group performance status (0 vs 1) and the baseline disease stage (recurrent vs primary metastatic).

[d] Confidence interval was computed using the Clopper-Pearson method.

[e] Computed using the Mantel-Haenszel method, stratified by the baseline Eastern Cooperative Oncology Group performance status (0 vs 1) and the baseline disease stage (recurrent vs primary metastatic).

[f] Computed from the Cochran-Mantel-Haenszel test, stratified by the baseline Eastern Cooperative Oncology Group performance status (0 vs 1) and the baseline disease stage (recurrent vs primary metastatic).

[g] Computed from the unstratified Cox proportional hazards regression model.

[h] Computed from the unstratified log-rank test.

[i] Confidence interval for the difference was computed using the Miettinen-Nurminen method.

[j] Computed from the Fisher exact test.

[k] Computed from the quantile regression model using the sparsity function.

anti-PD-1/PD-L1 therapy. Compared with the historical median overall survival of 22.1 months for gemcitabine-cisplatin, the median overall survival of the placebo plus gemcitabine-cisplatin group in the current study reached 33.7 (95% CI, 27.0-44.2) months, indicating a potential crossover effect on survival.

The subgroup analysis of the overall survival treatment effects demonstrated consistent results favoring toripalimab in majority of the subgroups, including PD-L1 expression and EBV copy number subgroups. One exception was patients with primary metastatic disease, with an HR of 1.23 (95% CI, 0.70-2.18) favoring placebo. We believe the overall survival treatment effect in patients with primary metastatic disease might be confounded by significant crossover and warrants careful interpretation. First, the subgroup analysis of progression-free survival, which was not affected by crossover, showed favorable toripalimab treatment effects in patients with primary metastatic disease (HR, 0.60 [95% CI, 0.37-0.98]). Second, a higher percentage of patients with primary metastatic NPC (42% [52/124]) received later-line

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Table 3. Treatment-Emergent Adverse Events Reported in at Least 20% of the Patients in Either Treatment Group

	Toripalimab + gemcitabine-cisplatin (n = 146)		Placebo + gemcitabine-cisplatin (n = 143)	
Adverse event, No. of patients (%)[a]	Any grade	≥Grade 3[b]	Any grade	≥Grade 3
Any treatment-emergent adverse event[c,d]	146 (100)	131 (89.7)	143 (100.0)	129 (90.2)
Leukopenia	133 (91.1)	90 (61.6)	135 (94.4)	84 (58.7)
Anemia	130 (89.0)	72 (49.3)	135 (94.4)	58 (40.6)
Neutropenia	126 (86.3)	86 (58.9)	133 (93.0)	91 (63.6)
Nausea	103 (70.5)	2 (1.4)	121 (84.6)	4 (2.8)
Vomiting	99 (67.8)	3 (2.1)	94 (65.7)	3 (2.1)
Thrombocytopenia	94 (64.4)	49 (33.6)	88 (61.5)	41 (28.7)
Decreased appetite	81 (55.5)	1 (0.7)	90 (62.9)	0
Constipation	58 (39.7)	0	66 (46.2)	0
Aspartate aminotransferase increased	58 (39.7)	2 (1.4)	45 (31.5)	2 (1.4)
Alanine aminotransferase increased	56 (38.4)	2 (1.4)	57 (39.9)	0
Fatigue	54 (37.0)	3 (2.1)	54 (37.8)	3 (2.1)
Hypothyroidism	53 (36.3)	1 (0.7)	25 (17.5)	0
Rash	51 (34.9)	5 (3.4)	39 (27.3)	3 (2.1)
Pyrexia	47 (32.2)	2 (1.4)	35 (24.5)	1 (0.7)
Diarrhea	45 (30.8)	2 (1.4)	33 (23.1)	0
Neuropathy peripheral	45 (30.8)	0	45 (31.5)	1 (0.7)
Cough	41 (28.1)	0	38 (26.6)	0
Hyponatremia	38 (26.0)	13 (8.9)	53 (37.1)	6 (4.2)
Musculoskeletal pain	37 (25.3)	0	36 (25.2)	1 (0.7)
Upper respiratory tract infection	37 (25.3)	5 (3.4)	20 (14.0)	4 (2.8)
Hypokalemia	35 (24.0)	12 (8.2)	36 (25.2)	11 (7.7)
Insomnia	35 (24.0)	0	26 (18.2)	0
Stomatitis	34 (23.3)	1 (0.7)	27 (18.9)	1 (0.7)
Dizziness	32 (21.9)	0	31 (21.7)	1 (0.7)
Blood creatinine increased	28 (19.2)	1 (0.7)	34 (23.8)	1 (0.7)
Headache	27 (18.5)	0	32 (22.4)	0
Hypochloremia	27 (18.5)	2 (1.4)	37 (25.9)	0

Data cutoff date: May 8, 2022.

[a] All adverse events were defined and graded according to the National Cancer Institute–Common Terminology Criteria for Adverse Events (NCI-CTCAE) version 5.0.

[b] An adverse event of grade 3 or higher per CTCAE (ie, severe, life-threatening or results in death).

[c] Treatment-emergent adverse events were all adverse events, regardless of relationship to the study drug, emerged after the study drug administration until 60 days after the last dose of the study drug or start of a new anticancer therapy, whichever occurred first.

[d] Refer to eTables 8-12 in Supplement 3 for additional information on adverse events.

anti-PD-1/PD-L1 therapy than patients with recurrent NPC (27% [45/165]), indicating a potential impact of imbalanced crossover on the primary metastatic subgroup.

Because latent EBV infection is critical to the development of NPC, the dynamic change of plasma EBV DNA copy number is closely related with response to chemotherapy, radiotherapy,[18,19] or immunotherapy.[15] In the current study, more patients from the toripalimab group experienced EBV DNA copy number reduction to undetectable level compared with the placebo group. In addition, significantly fewer patients experienced EBV DNA copy number rebound in the toripalimab group, consistent with higher percentage of toripalimab-treated patients achieving long-term clinical benefits. Notably, the rebound preceded investigator-assessed disease progression by a median of 1.9 months in the toripalimab group, suggesting that EBV DNA copy number rebound might be used to predict disease progression.

The safety profile of toripalimab combined with chemotherapy observed in this study was largely consistent with the profile of chemotherapy alone and with results from other first-line toripalimab-chemotherapy combination studies.[12,20] Compared with the interim results, no novel adverse events emerged after extended exposure to toripalimab for up to 2 years.[10]

Limitations

First, because age is not one of the stratification factors used for randomization in this study, an imbalance of age group distribution (≤50 vs >50 years) between the 2 groups occurred. Compared with the placebo group, the median age in the toripalimab group was 5 years younger, and nearly 20% more of the patients were 50 years or younger. By conducting a multivariate Cox regression model with age (≤50 vs >50 years) as a covariate, the adjusted HR was 0.66 (95% CI, 0.46-0.94) for overall survival and 0.53 (95% CI, 0.38-0.74) for progression-free survival. The results were consistent with the primary analysis for overall survival and progression-free survival.

Second, single-agent gemcitabine or capecitabine, while not recommended by practice guidelines in the US or EU, is commonly used as a maintenance therapy after the initial gemcitabine-cisplatin doublet treatment for RM-NPC. In hindsight, the ideal study design would have compared toripalimab vs single-agent chemotherapy or toripalimab plus chemotherapy vs placebo plus chemotherapy during the

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maintenance phase. Since anti-PD-1 therapy has a more favorable safety profile compared with chemotherapy, this design would have allowed risk benefit assessments to determine the optimal maintenance regimen.

Third, the current study exclusively enrolled patients in NPC-endemic Asian regions, where the majority of patients have nonkeratinizing histology, closely associated with EBV infection. In contrast, keratinizing NPC, which accounts for 25% of NPC in the West, is not caused by EBV infection. The subgroup analysis demonstrated that the treatment effects of overall survival favored toripalimab in both the subgroup with high EBV copy number and the subgroup with low EBV copy number, indicating that toripalimab will benefit patients with low EBV copy numbers, who are representative of patients with NPC in the West.

Conclusions

The addition of toripalimab to chemotherapy as first-line treatment for RM-NPC provided statistically significant and clinically meaningful progression-free survival and overall survival benefits compared with chemotherapy alone, with a manageable safety profile. These findings support the use of toripalimab plus gemcitabine-cisplatin as the new standard of care for this patient population.

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